

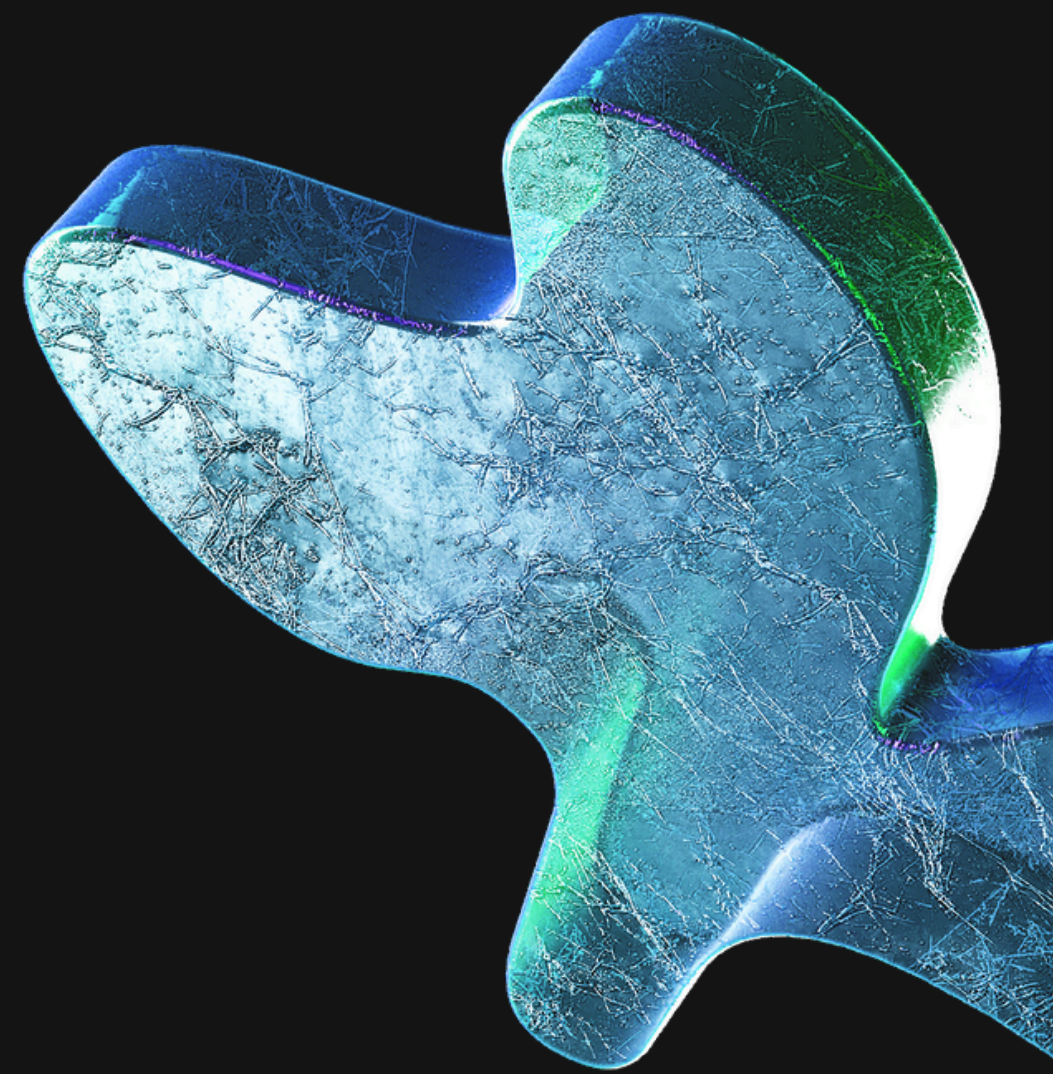


# Automated cryptocurrency trading system with a Telegram bot

## C-BOT

Automated, convenient, fast, and high-quality  
trading

(English Translation by Onyshchuk Bohuslav)



In this presentation, we will discuss the development and testing of software for automated cryptocurrency trading that integrates with the Telegram platform. We will examine the types of trades, how the program's algorithm works, the integration of the program into Telegram, and the advantages and disadvantages of using the program.

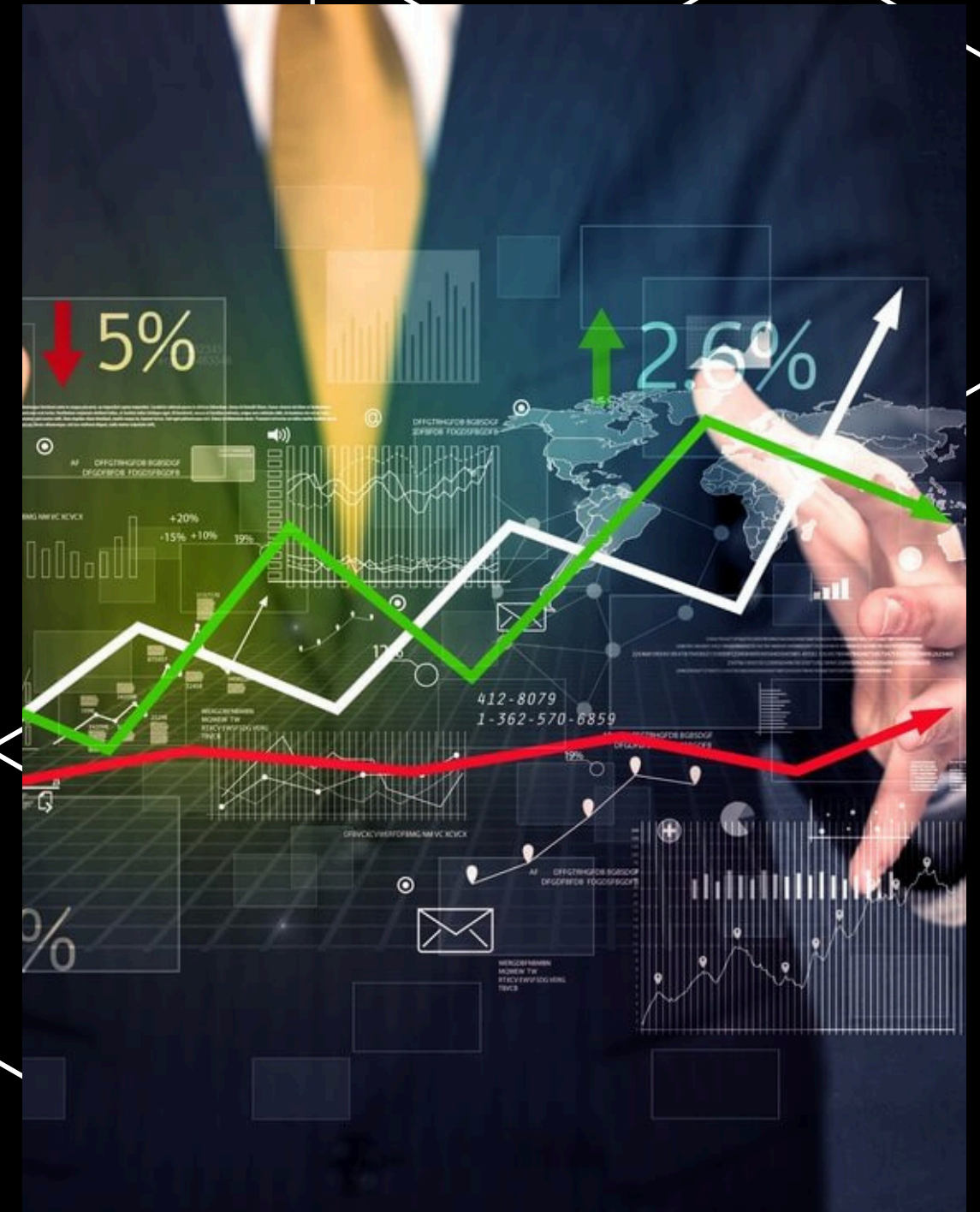
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# Introduction

- In today's world, cryptocurrencies are becoming increasingly popular as an investment asset.
- Their volatility creates opportunities for profit, but also carries risks.
- Automated trading using software can help traders make informed decisions and minimize risks.



# Research Objectives

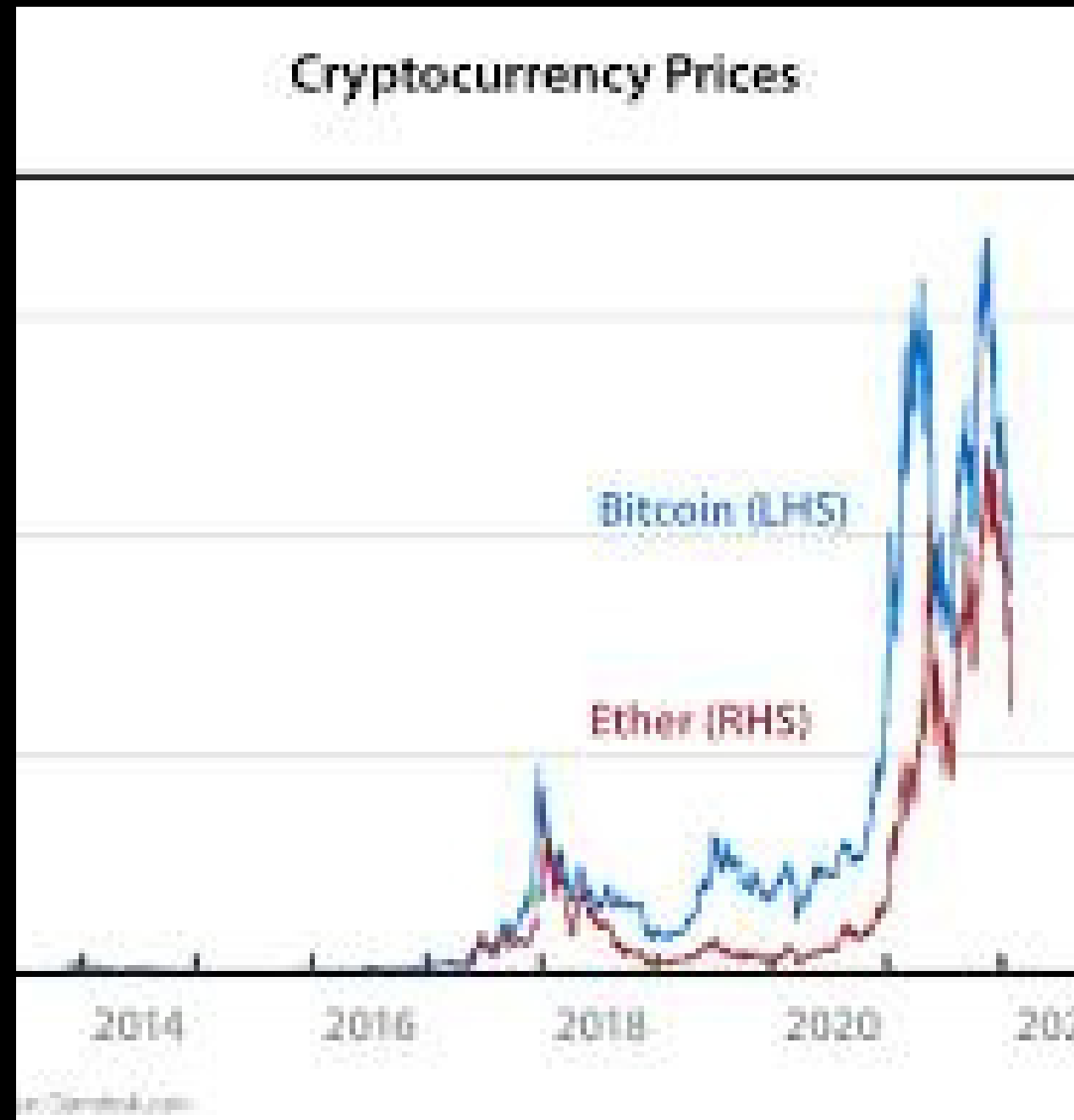


- Research and categorize the various types of transactions in the cryptocurrency market.
- Develop a cryptocurrency trading algorithm based on technical analysis.
- Implement the ability to interact with the developed software via a Telegram bot.
- Test and evaluate the effectiveness of the developed software.



# Subject of the study

Software designed for automated cryptocurrency trading in the financial market.



# Subject of the study

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Integration and automation of trading algorithms into software.

# Practical significance

**1** The software we have developed can help traders:

**2** Improve trade efficiency.

**3** Minimize risks.

**4** Save time.

**5** Make informed decisions based on technical analysis.





# Chapter 1: Types of Transactions and How the developed Program's Algorithm Works

- 1.1 Overview of Types of Exchange Trades
- 1.2 Technical Analysis and How It Is Used in the Program
- 1.3 Program Algorithm





# 1.1 Overview of Types of Exchange Transactions

Market order: immediate execution at the current market price.

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Limit order: executed at the specified price or better.

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Stop-loss order: automatic closure of a position when a certain loss level is reached.

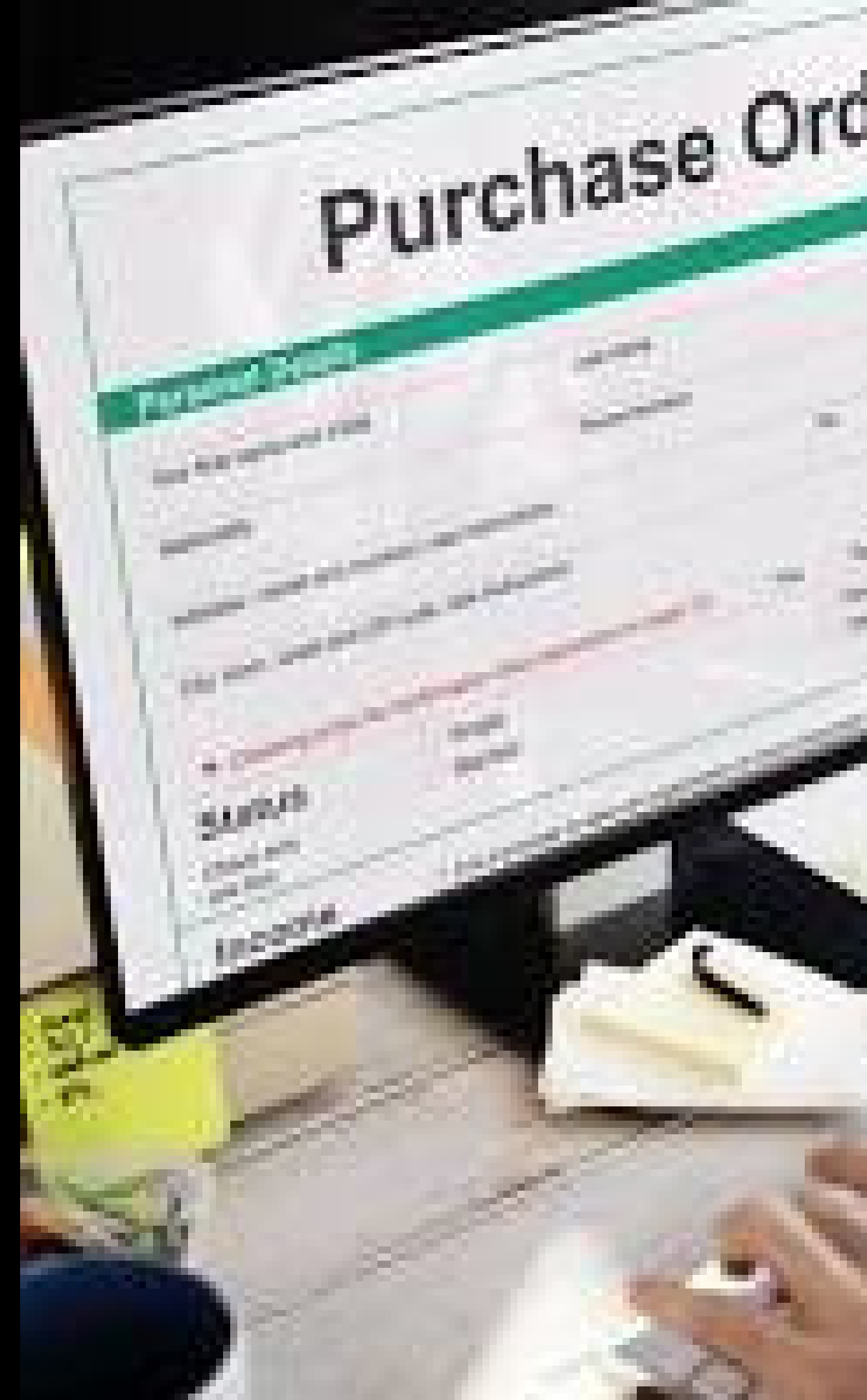
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Take-profit order: automatic closure of a position when a certain profit level is reached.

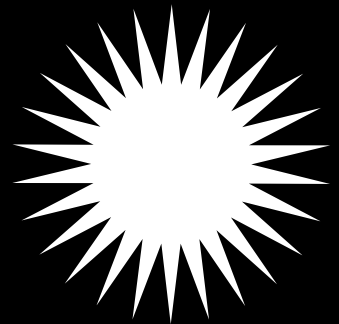
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Conditional orders: orders that are executed under certain conditions, such as when a specific price level or indicator is reached.

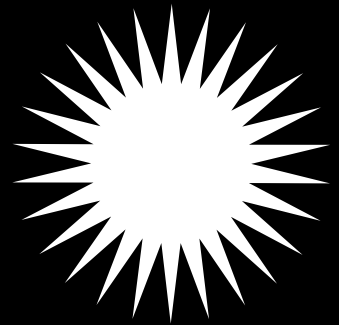
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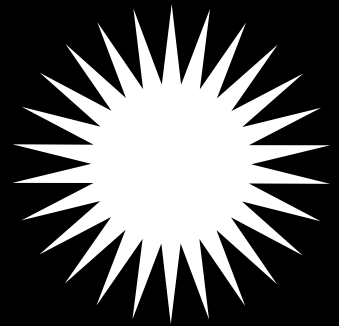
# 1.2 Technical Analysis and How It Is Used in the Program



The program uses technical analysis to identify trends and patterns in the cryptocurrency market.



She uses technical indicators such as the RSI, MACD, stochastic oscillator, and others.



The program decides when to open and close positions based on signals received from technical indicators.



# 1.3 Program Algorithm

- The program retrieves price data from the Binance cryptocurrency exchange.
- It analyzes technical indicators for various time frames.
- It makes decisions about opening and closing positions based on the signals received.
- It automatically creates buy and sell orders for cryptocurrencies.
- It monitors positions and closes them when stop-loss or take-profit orders are triggered.

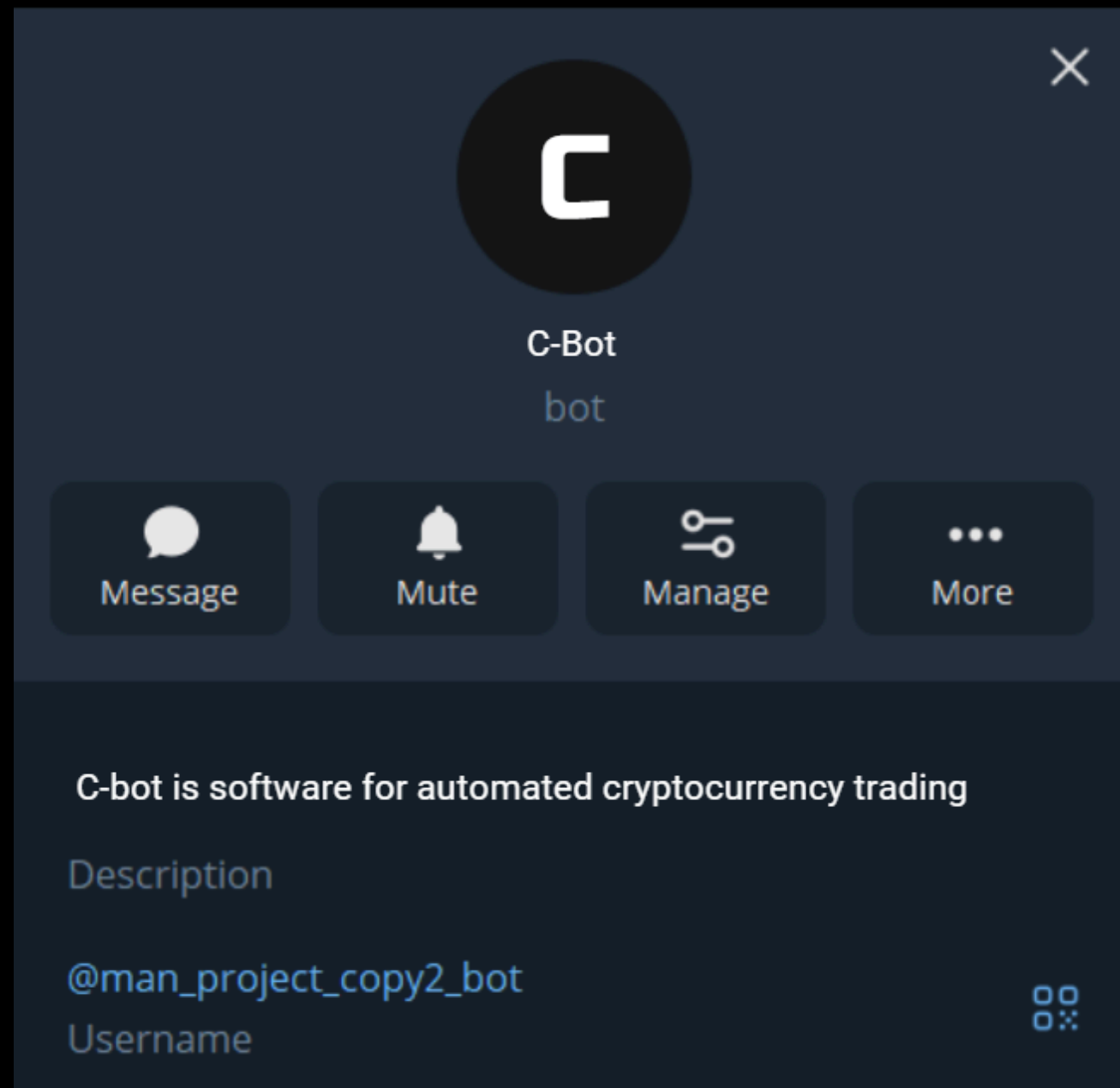
```
def top_coin():  
    all_tickers = pd.DataFrame(client.get_ticker())  
    usdt = all_tickers[all_tickers.symbol.str.contains('USDT')]  
    work = usdt[~((usdt.symbol.str.contains('UP')) | (usdt.symbol.str.contains('DOWN')))]  
    top_coin = work[work.priceChangePercent == work.priceChangePercent.max()]  
    top_coin = top_coin.symbol.values[0]  
    return top_coin  
  
inter = {1: Interval.INTERVAL_1_MINUTE,  
         15: Interval.INTERVAL_15_MINUTES,  
         30: Interval.INTERVAL_30_MINUTES,  
         60: Interval.INTERVAL_1_HOUR}
```

# Chapter 2: Integrating the App into Telegram and How It Works

- 2.1 Integrating the app into Telegram
- 2.2 The app's features in Telegram
- 2.3 Pros and cons of using the app



## 2.1 Integrating the app into Telegram

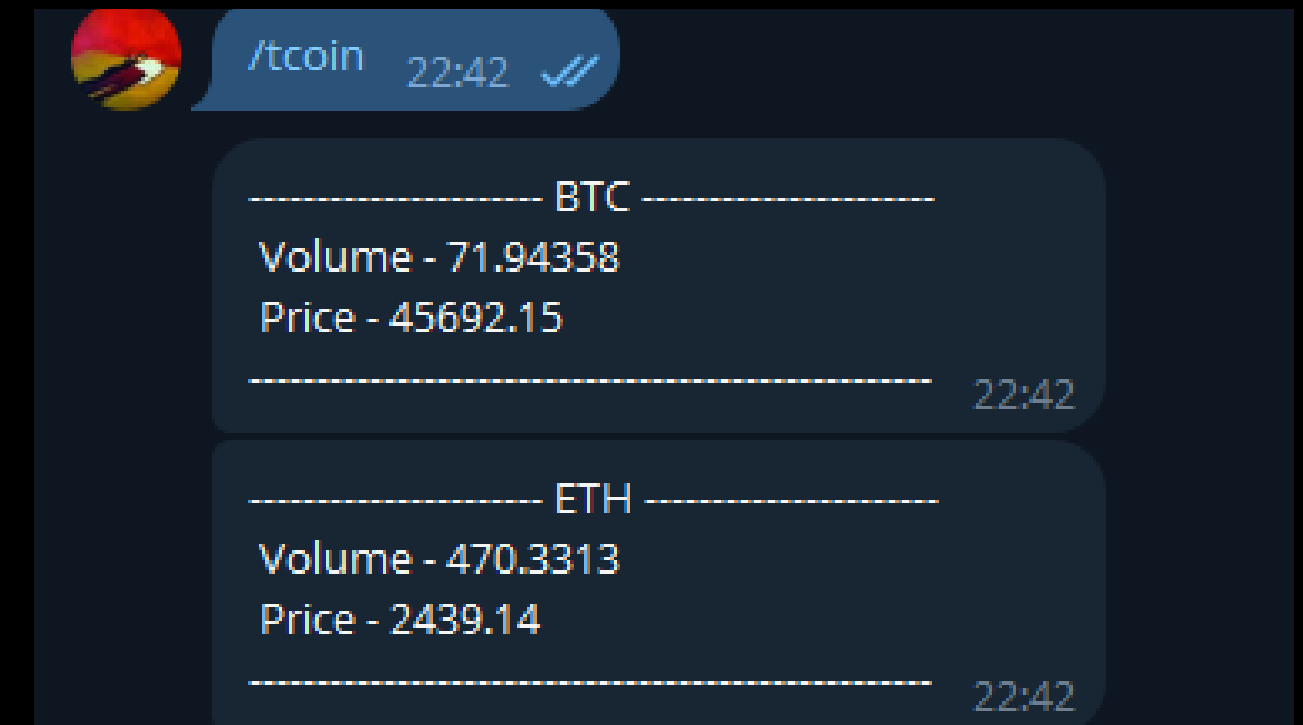


- The program integrates with Telegram using the Telegram API.
- A Telegram bot is created to serve as an interface for interacting with the program.
- Users can control the program and retrieve information using the Telegram bot's commands.

## 2.2 The app's features in Telegram

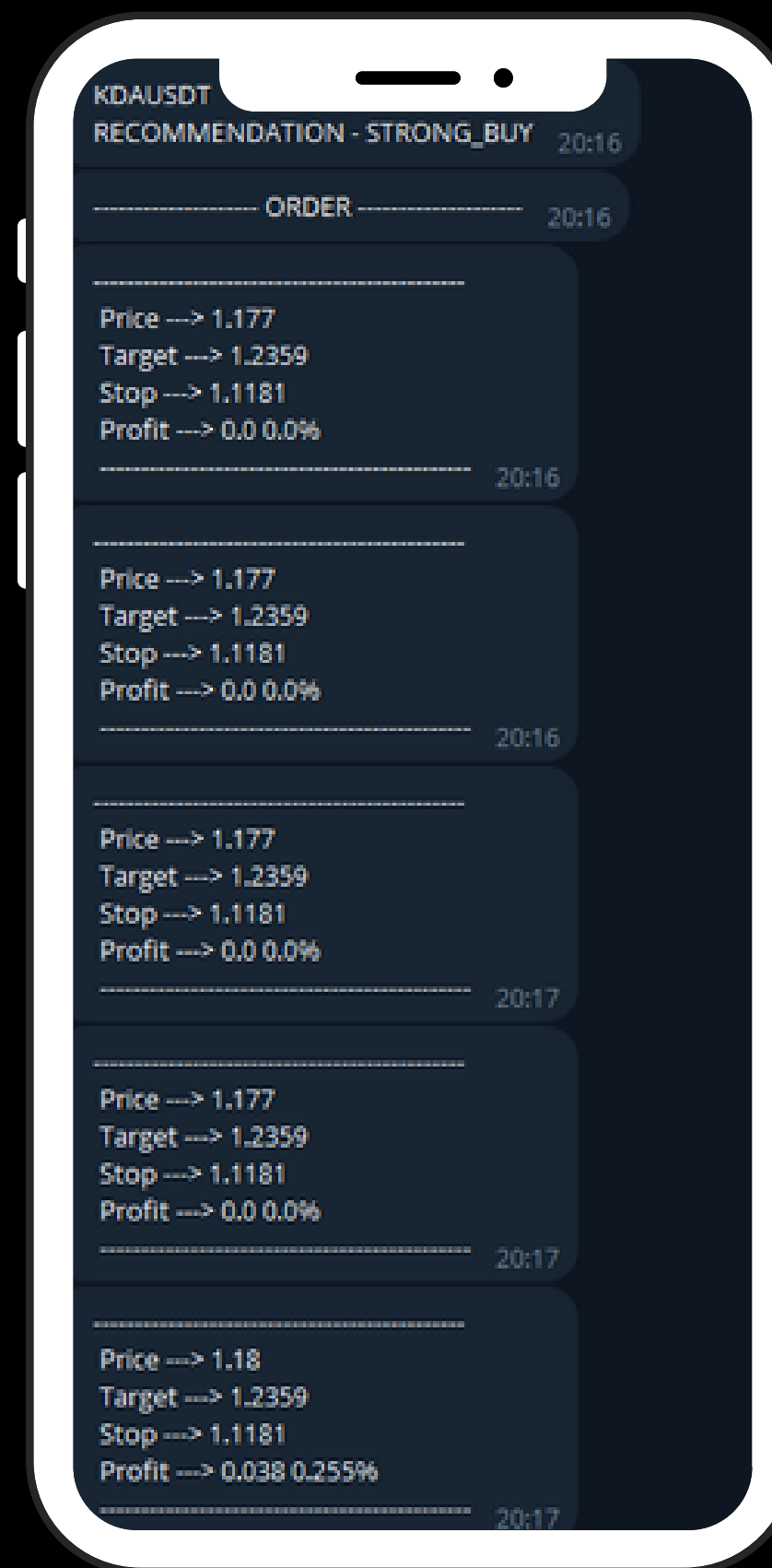
**Users can get information about the state of the cryptocurrency market.**

- Users can receive trading recommendations.
- Users can manage their trading positions.
- Users can view statistics and results of their trades.



## 2.3 Переваги та недоліки використання програми

- Збільшення ефективності торгівлі.
- Зменшення ризиків.
- Економія часу.
- Прийняття обґрунтованих рішень на основі технічного аналізу.
- Доступність 24/7.



# Conclusion

Automated cryptocurrency trading can be an effective tool for traders who want to increase their efficiency and reduce risks.

The program developed as part of this study can help traders make informed decisions based on technical analysis and automate their trading operations.

It is important to remember that automated trading does not guarantee profits, and traders should always use appropriate risk management strategies.

